ENIGMA DARK

Securing the Shadows

Security Review

Euler

EVK Periphery

MigrationHelper

Jul, 2026

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Summary

Enigma Dark
Enigma Dark is a web3 security firm leveraging the best talent in the space to secure all kinds of blockchain protocols and decentralized apps. Our team comprises experts who have honed their skills at some of the best auditing companies in the industry. With a proven track record as highly skilled white-hats, they bring a wealth of experience and a deep understanding of the technology and the ecosystem.

Learn more about us at enigmadark.com

Euler evk-periphery Migration Helper
The Euler evk-periphery Migration Helper introduces the `MigrationHelper` contract, inherited by the `SwapVerifier`, which provides from-sender primitives for migrating lending positions in and out of Euler. Operating within an Ethereum Vault Connector (EVC) batch, it pulls tokens and performs debt- and collateral-bearing actions on external lending protocols (Aave V3 and Morpho Blue) on behalf of the EVC-authenticated sender, enabling capital-free migrations of leveraged positions between Aave, Morpho, and Euler vaults.

Engagement Overview

Over the course of 1 day, in July 2026, the Enigma Dark team conducted a security review of the Euler evk-periphery Migration Helper. The review was performed by two Lead Security Researchers: vnmrtz & kiki.

The following repositories were reviewed at the specified commits:

Repository	Commit
evk-periphery	`diff ad5325fe...469d9cfe` (PR #423)

Scope

The review was limited to the changes between the above commits, covering the following contracts under `src/`:

```
src/
└── Swaps/
    ├── MigrationHelper.sol
    └── SwapVerifier.sol
```

Risk Classification

Severity	Description
Critical	Vulnerabilities that lead to a loss of a significant portion of funds of the system.
High	Exploitable, causing loss or manipulation of assets or data.
Medium	Risk of future exploits that may or may not impact the smart contract execution.
Low	Minor code errors that may or may not impact the smart contract execution.
Informational	Non-critical observations or suggestions for improving code quality, readability, or best practices.

Vulnerability Summary

Severity	Count	Fixed	Acknowledged
Critical	0	0	0
High	0	0	0
Medium	0	0	0
Low	0	0	0
Informational	0	0	0

Findings

High Risk

No issues found.

Medium Risk

No issues found.

Low Risk

No issues found.

Informational

No issues found.

Annex: Documented Assumptions & References

The migration primitives depend on the following assumptions, documented in the codebase:

1. The privileged `*ForSender` legs (`aaveBorrowForSender`, `morphoBorrowForSender`, `morphoWithdrawCollateralForSender`) must be top-level EVC batch items with `onBehalfOfAccount` set to the user, not nested inside `Swapper.multicall`; otherwise `_msgSender()` does not resolve to the user and the call reverts.
2. EVC operators are trusted by the account owner and can exercise any external authorization left standing on this contract.
3. Aave credit delegations and Morpho operator authorizations should be granted and revoked within the migration batch (`delegationWithSig` / `setAuthorizationWithSig`), not left standing.
4. The contract holds no funds between calls and is intended for use within a single atomic batch.

References: `docs/swaps.md` (Position migration helpers), `src/Swaps/MigrationHelper.sol` NatSpec, and the Ethereum Vault Connector documentation.

Disclaimer

This report does not endorse or critique any specific project or team. It does not assess the economic value or viability of any product or asset developed by parties engaging Enigma Dark for security assessments. We do not provide warranties regarding the bug-free nature of analyzed technology or make judgments on its business model, proprietors, or legal compliance.

This report is not intended for investment decisions or project participation guidance. Enigma Dark aims to improve code quality and mitigate risks associated with blockchain technology and cryptographic tokens through rigorous assessments.

Blockchain technology and cryptographic assets inherently involve significant risks. Each entity is responsible for conducting their own due diligence and maintaining security measures. Our assessments aim to reduce vulnerabilities but do not guarantee the security or functionality of the technologies analyzed.

This security engagement does not guarantee against a hack. It is a review of the codebase during a specific period of time. Enigma Dark makes no warranties regarding the security of the code and does not warrant that the code is free from defects. By deploying or using the code, the project and users of the contracts agree to use the code at their own risk. Any modifications to the code will require a new security review.